

Executive Summary: APL Logistics - Delivery Performance and Delay Risk Analysis

Overview

This project evaluates delivery performance across APL Logistics' global fulfillment network using 180,519 validated shipment records. The objective is to identify delay exposure, compare shipping-mode effectiveness, and pinpoint the regional and market conditions that most strongly affect service reliability.

Key Findings

- Only 42.7% of shipments were delivered on time or early, while 57.3% were delayed.
- Standard Class delivered the strongest results, with a 60.2% on-time rate and the highest logistics-efficiency score.
- First Class and Second Class showed the weakest outcomes, with materially higher late-delivery risk and larger delivery gaps.
- Central Africa was the highest-risk region, Europe was the weakest market, and Home Office customers showed the highest segment-level delay exposure.

Operational Impact

The results indicate that delivery risk is not isolated to a few exceptional orders. It reflects a broader service-performance issue that can affect SLA compliance, customer confidence, and operating cost through escalations, rework, and preventable delay management.

Data Limitation

The source dataset does not contain date-related columns for order creation, dispatch, or shipment completion. As a result, the current analysis supports operational benchmarking but cannot provide time-trend, seasonality, or aging analysis.

Recommendations

- Audit First Class and Second Class service design, carrier allocation, and execution quality.
- Prioritize corrective route interventions in Central Africa and elevated-risk European corridors.
- Use Standard Class operating practices as a benchmark for weaker shipping modes.
- Enrich the source system with real order and shipment date fields so future analytics can support time-based logistics governance.

Conclusion

The Project 4 dashboard and research paper provide a strong decision-support baseline for monitoring logistics performance. With targeted action on high-risk modes and geographies, APL Logistics can improve delivery reliability and strengthen operational transparency.